

Modelos ARIMA/SARIMA

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Setup

```
library(tseries)

## Registered S3 method overwritten by 'quantmod':
##   method      from
##   as.zoo.data.frame zoo

library(TTR)
library(forecast)
library(dotenv) # Automatically reads .env file
source("inegi.R")

##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##   filter, lag

## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union

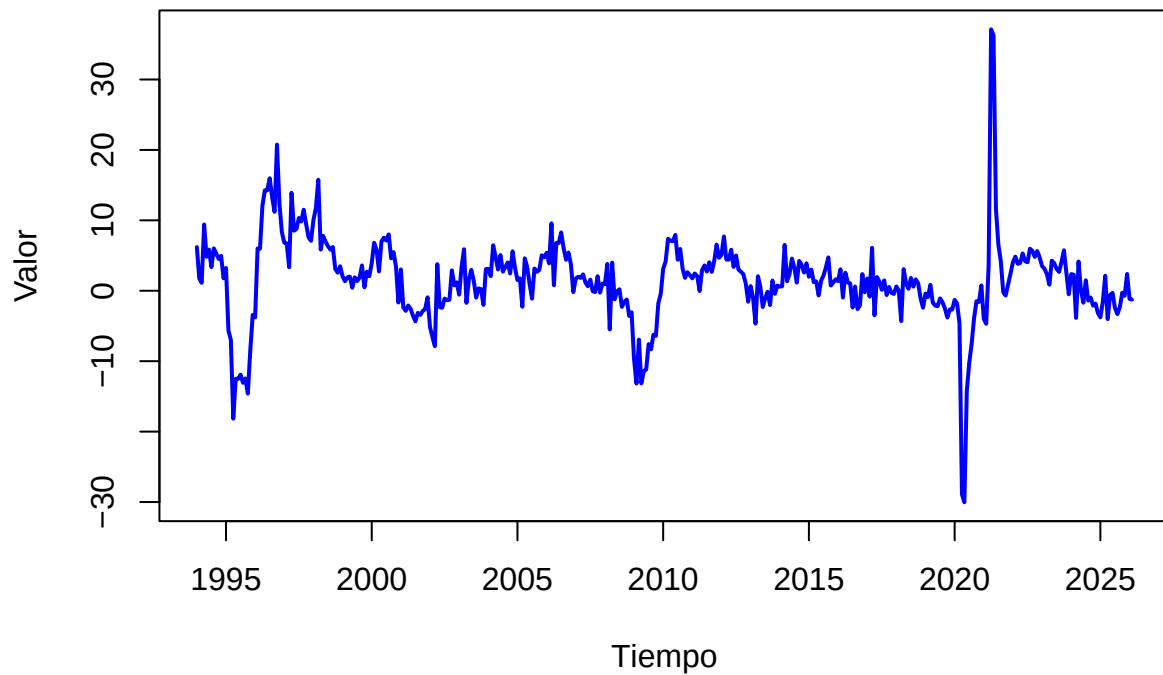
readRenviron(".env")

token <- Sys.getenv("INEGI_TOKEN")
st1.id <- "736526"
st2.id <- "735879"
```

Primera Serie

```
st1.ts <- datos_inegi_total(st1.id, token)
```

Indicador: 736526



```
## --- Resumen de la Serie ---  
## Inicio detectado: 1994 / 1  
## Frecuencia: 12  
## Registros: 386
```

Modelo ARIMA

```
adf.test(st1.ts, k=12)
```

```
## Warning in adf.test(st1.ts, k = 12): p-value smaller than printed p-value
```

```
##
```

```
## Augmented Dickey-Fuller Test
```

```
##
```

```
## data: st1.ts
```

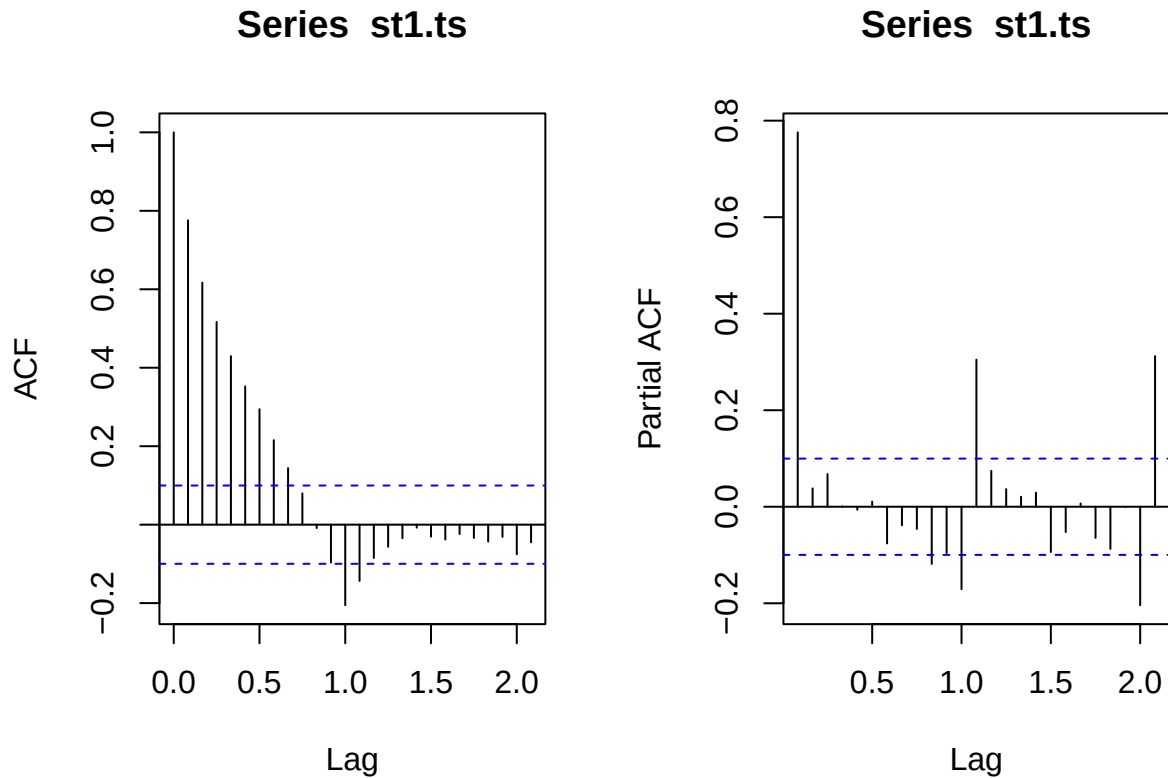
```
## Dickey-Fuller = -4.6716, Lag order = 12, p-value = 0.01
```

```
## alternative hypothesis: stationary
```

```
par(mfrow=c(1,2))
```

```
acf(st1.ts)
```

```
pacf(st1.ts)
```



Para $\alpha = 0.05$, como $\alpha > p$ podemos rechazar H_0 y por lo tanto, hay evidencia suficiente de que la serie es estacionaria.

Como la gráfica de la función ACF es decreciente, $q = 0$. Mientras que la de PACF se corta y por tanto, $p = 1$. Se propone el modelo $ARIMA(1, 0, 0)$

```
st1.arima <- Arima(st1.ts, order = c(1, 0, 0))
```

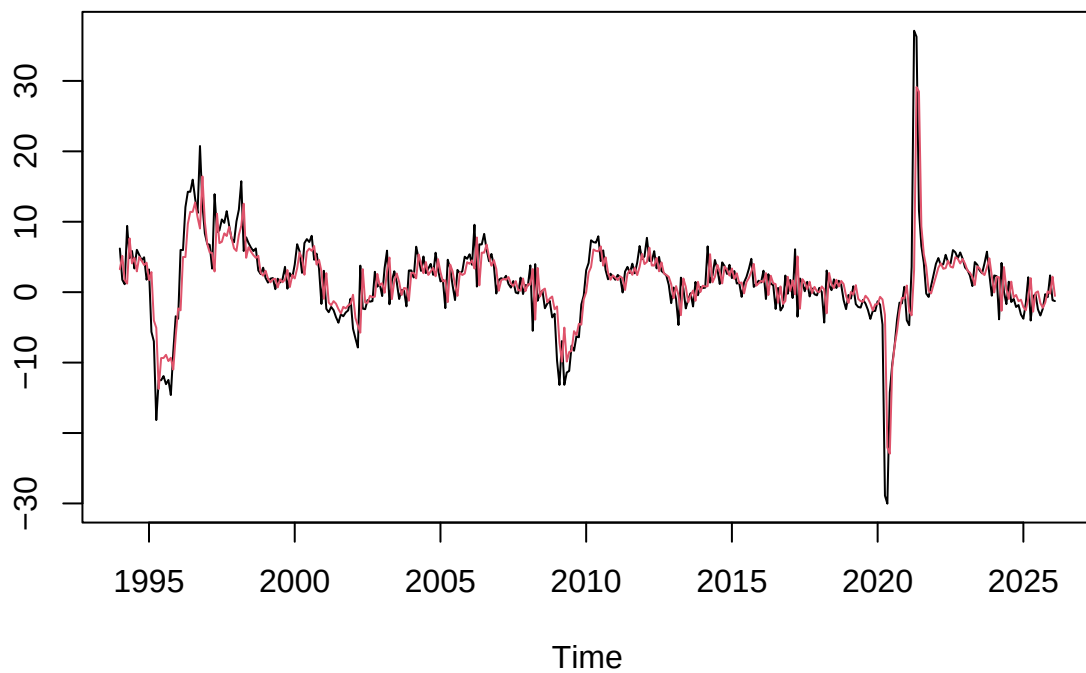
```
st1.arima
```

```
## Series: st1.ts
## ARIMA(1,0,0) with non-zero mean
##
## Coefficients:
##      ar1      mean
##      0.7754  1.5473
## s.e.  0.0319  0.8482
##
## sigma^2 = 14.33: log likelihood = -1060.98
## AIC=2127.95  AICc=2128.01  BIC=2139.82
```

```
summary(auto.arima(st1.ts, ic="bic"))
```

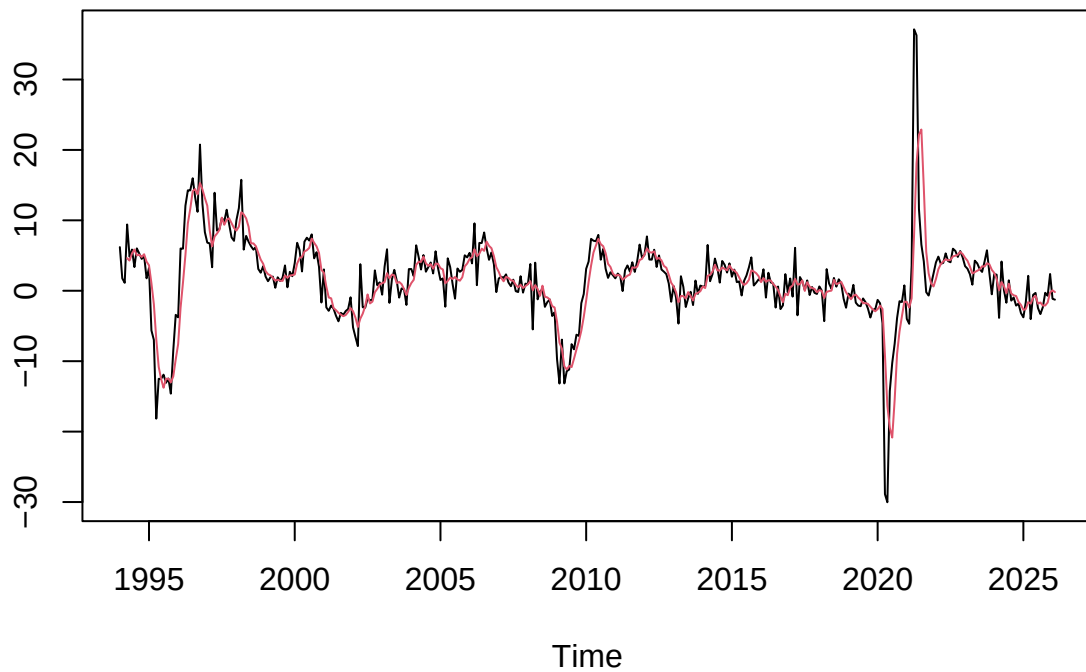
```
## Series: st1.ts
## ARIMA(2,0,1)(2,0,2)[12] with zero mean
##
## Coefficients:
##      ar1      ar2      ma1      sar1      sar2      sma1      sma2
##      1.6498 -0.6550 -0.8572 -0.5466 -0.0570 -0.0865 -0.4826
```

```
## s.e.  0.1288  0.1236  0.0983  0.1519  0.0879  0.1494  0.1268
##
## sigma^2 = 10.16:  log likelihood = -996.69
## AIC=2009.37  AICc=2009.76  BIC=2041.02
##
## Training set error measures:
##              ME      RMSE      MAE      MPE      MAPE      MASE      ACF1
## Training set 0.1975719 3.158899 2.088474 54.25757 132.1543 0.3657779 0.02231648
ts.plot(st1.ts, st1.arima$fitted, col=1:2)
```



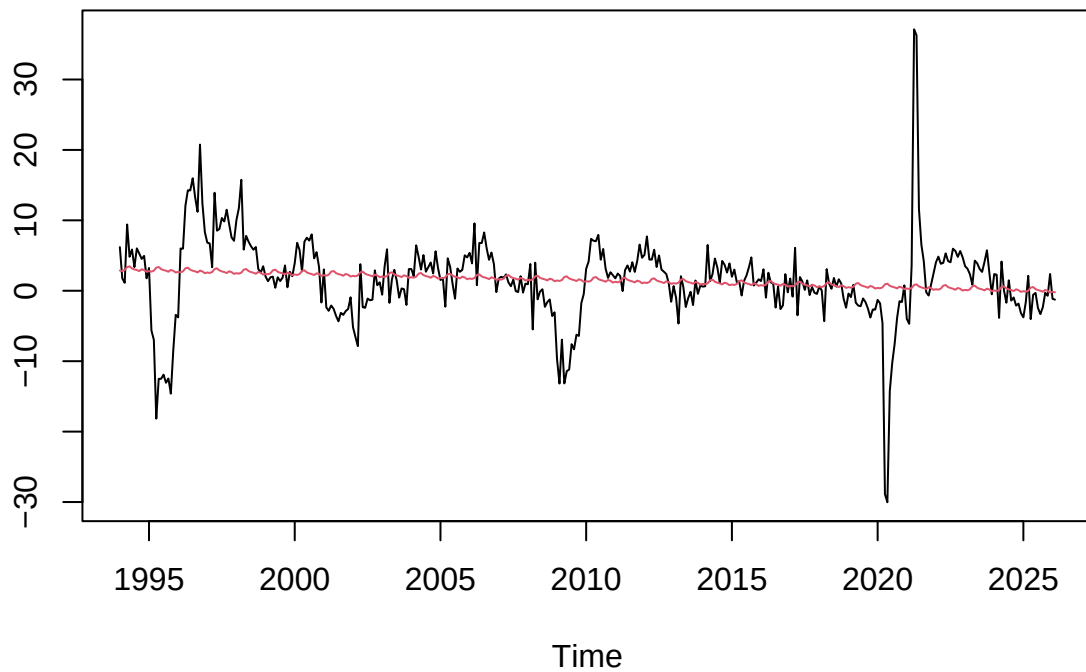
Modelo SMA

```
st1.sma <- SMA(st1.ts, n=4)
ts.plot(st1.ts, st1.sma, col=1:2)
```



Modelo MCS

```
st1.dummies <- seasonaldummy(st1.ts)
st1.mcs <- tslm(st1.ts ~ trend + season)
ts.plot(st1.ts, st1.mcs$fitted.values, col=1:2)
```



Pronósticos

Tenemos los pronósticos:

```
st1.arima.forecast <- forecast(st1.arima, h=6, level=95)
st1.sma.forecast <- forecast(st1.sma, h=6, level=95)
```

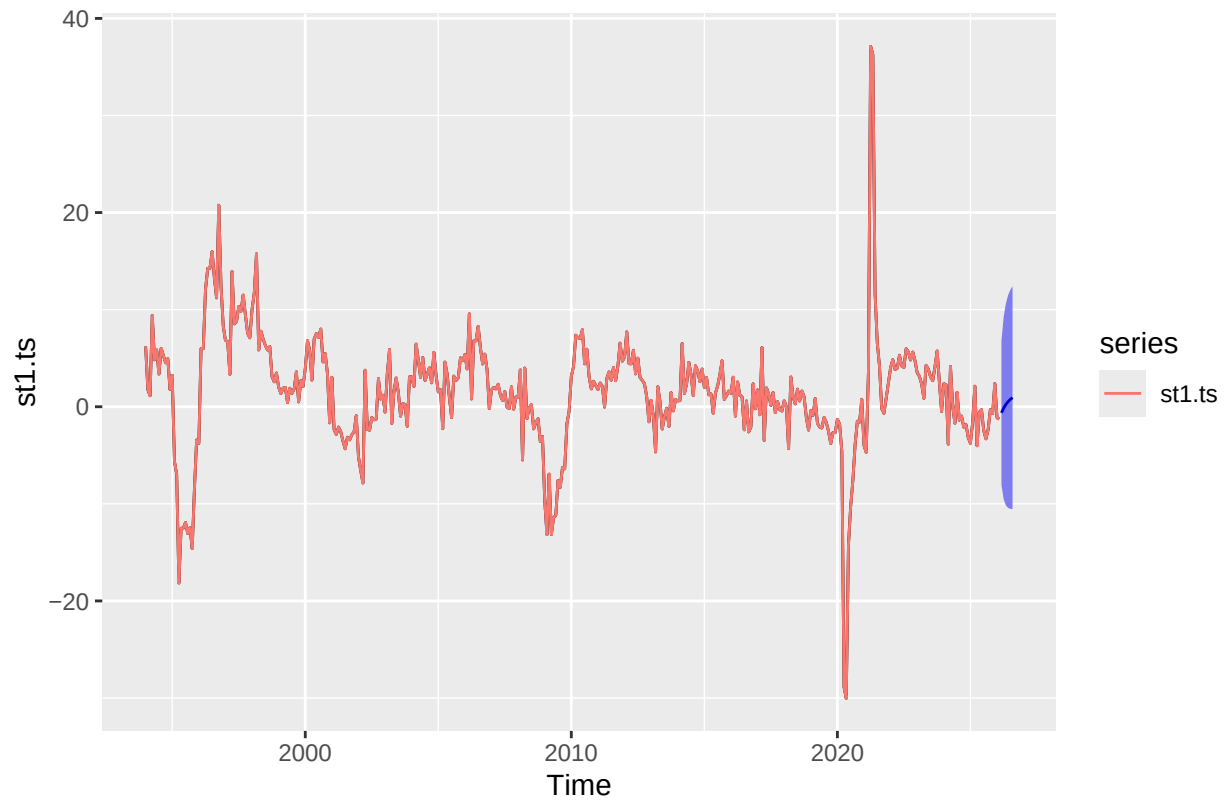
```
## Warning in lsfit(seq(maxn), head(y.sa, maxn)): 3 missing values deleted
## Warning in lsfit(seq(maxn), head(y.sa, maxn)): 3 missing values deleted
```

```
st1.mcs.forecast <- forecast(st1.mcs, h=6, level=95)
```

Comparación visual

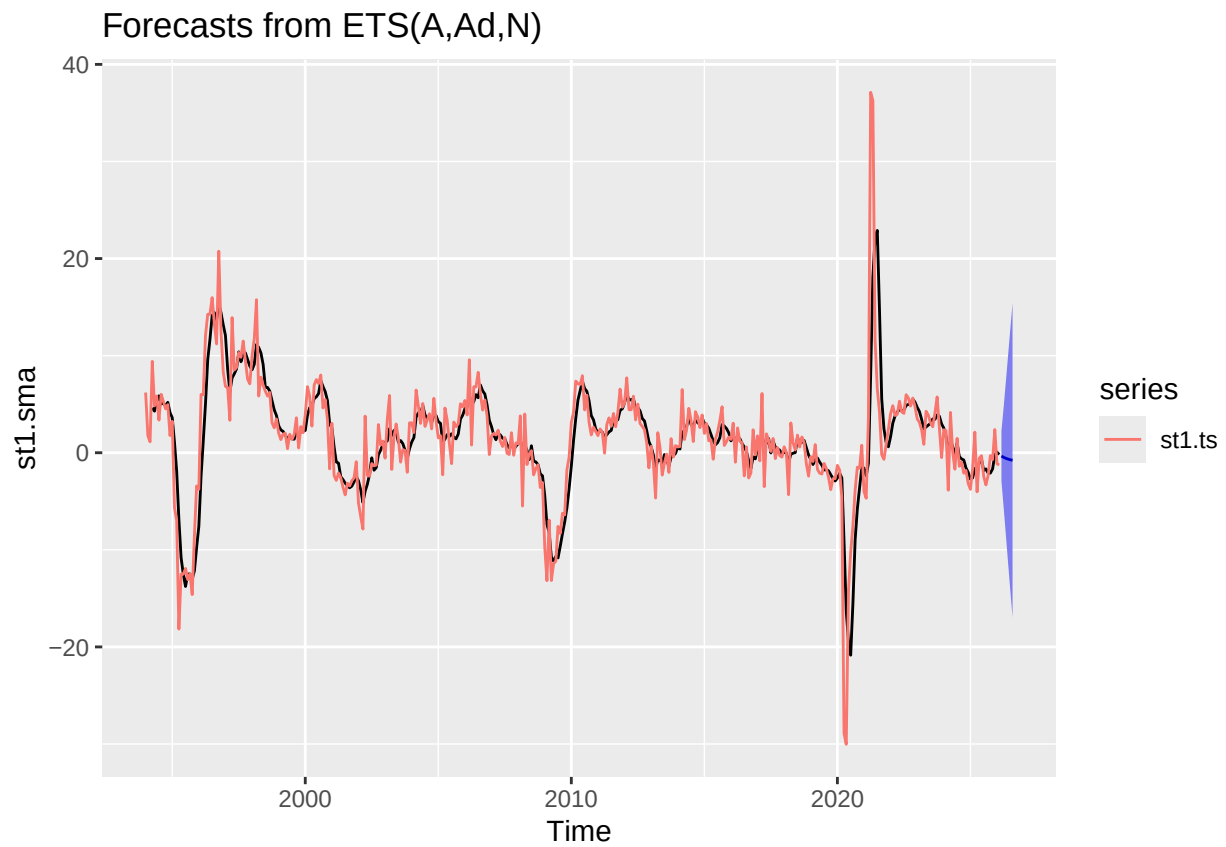
```
autoplot(st1.arima.forecast) + autolayer(st1.ts)
```

Forecasts from ARIMA(1,0,0) with non-zero mean

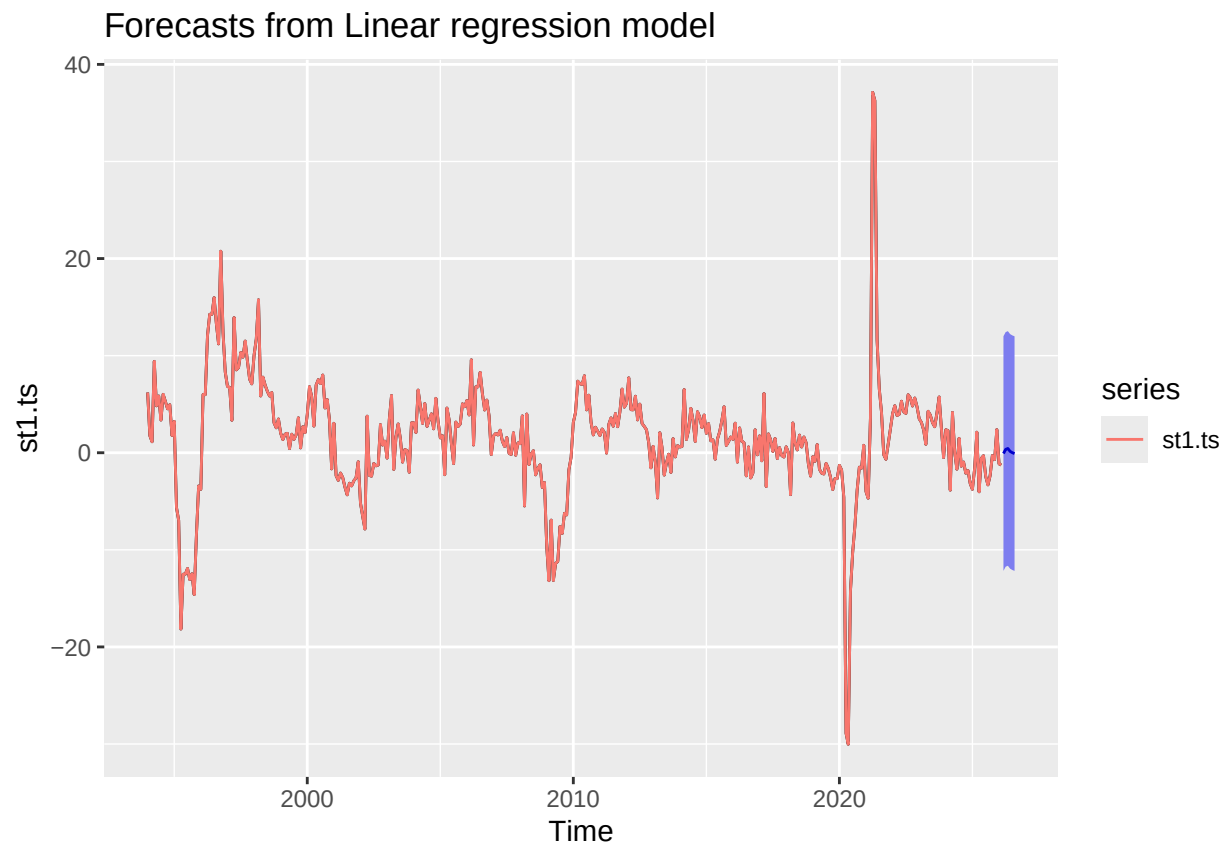


```
autoplot(st1.sma.forecast) + autolayer(st1.ts)
```

```
## Warning: Removed 3 rows containing missing values or values outside the scale range
## (`geom_line()`).
```



```
autoplot(st1.mcs.forecast) + autolayer(st1.ts)
```

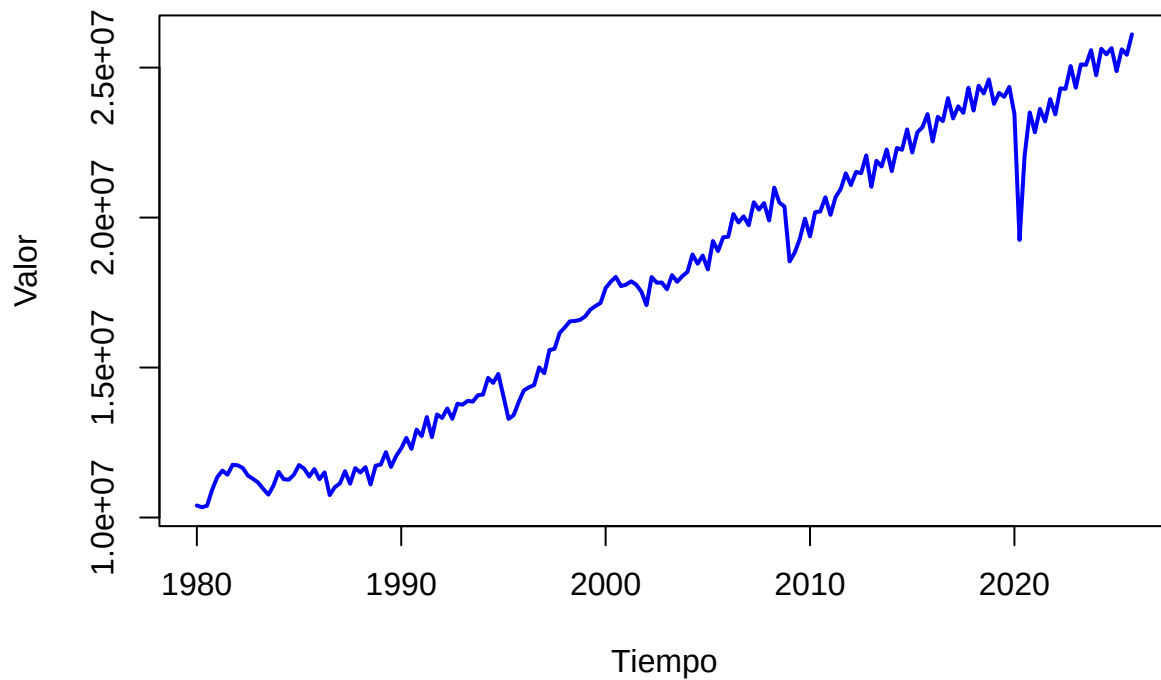
En este caso, parece que el modelo SMA da un mejor predicción aunque las estimaciones no son las mejores. Los modelos ARIMA y MCS parecen dar mejores estimaciones y predicciones similares.

Segunda serie

Modelo ARIMA

```
st2.ts <- datos_inegi_total(st2.id, token)
```

Indicador: 735879



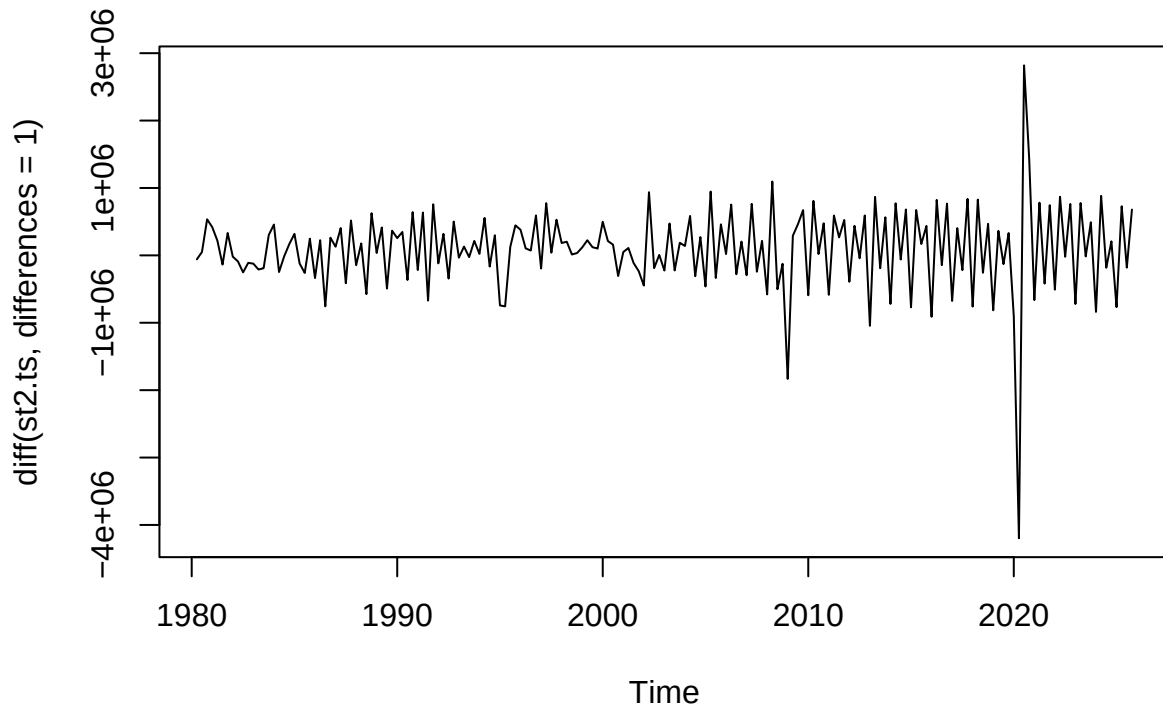
```
## --- Resumen de la Serie ---  
## Inicio detectado: 1980 / 1  
## Frecuencia: 4  
## Registros: 184
```

```
adf.test(st2.ts, k = 4)
```

```
##  
## Augmented Dickey-Fuller Test  
##  
## data: st2.ts  
## Dickey-Fuller = -3.4092, Lag order = 4, p-value = 0.05496  
## alternative hypothesis: stationary
```

Para $\alpha = 0.05$, como $\alpha < \text{p-value}$, no podemos rechazar H_0 . Intentamos con las diferencias.

```
plot(diff(st2.ts, differences=1))
```



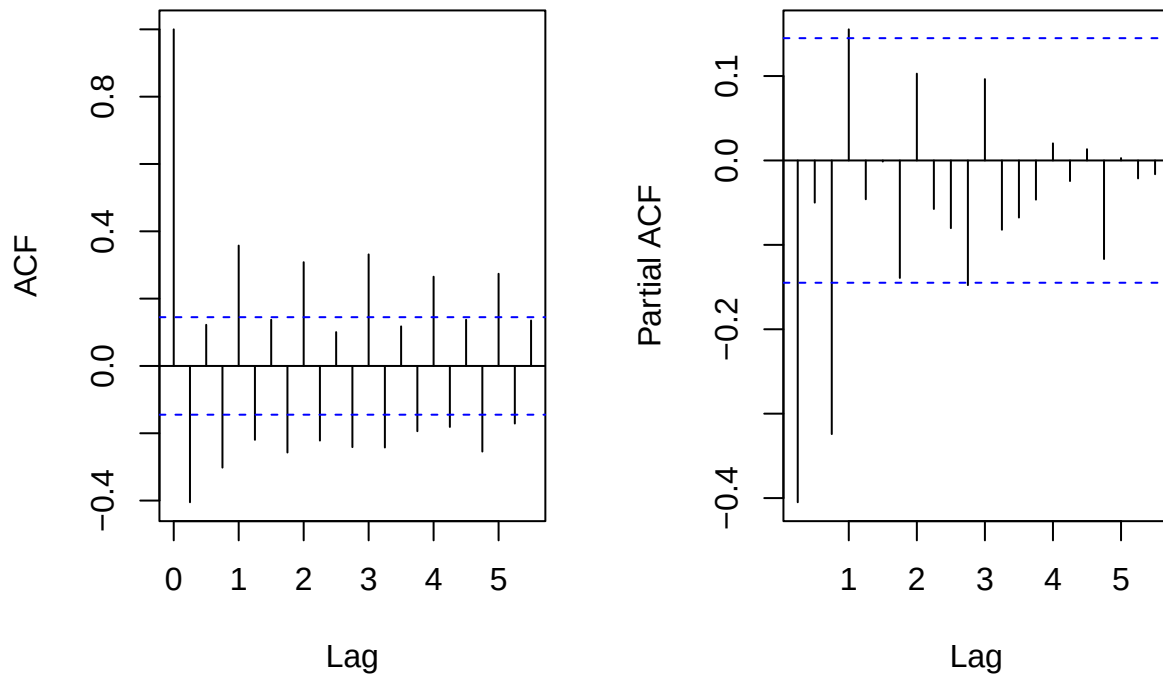
```
adf.test(diff(st2.ts), k=4)
```

```
## Warning in adf.test(diff(st2.ts), k = 4): p-value smaller than printed p-value
##
## Augmented Dickey-Fuller Test
##
## data: diff(st2.ts)
## Dickey-Fuller = -6.9498, Lag order = 4, p-value = 0.01
## alternative hypothesis: stationary
```

Esta vez, si hay evidencia suficiente para rechazar H_0 y podemos seguir con los siguientes pasos.

```
par(mfrow=c(1,2))
acf(diff(st2.ts, differences=1))
pacf(diff(st2.ts, differences=1))
```

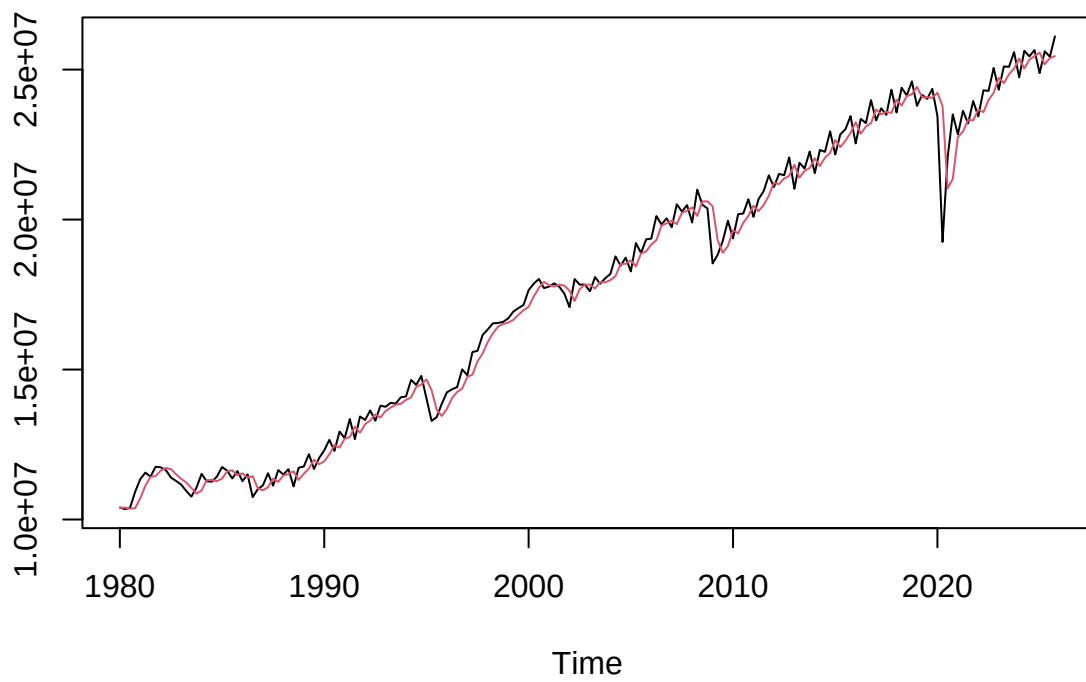
Series diff(st2.ts, differences = 1) Series diff(st2.ts, differences = 1)



Como fue necesario usar las diferencias, $d = 1$. Ya que en los correlogramas, ACF se corta $q = 1$ de igual forma que PCAF, i.e, $p = q$. Por lo tanto, se propone el modelo ARIMA(1, 1, 1).

```
st2.arima <- Arima(st2.ts, order = c(1, 1, 1))
st2.arima

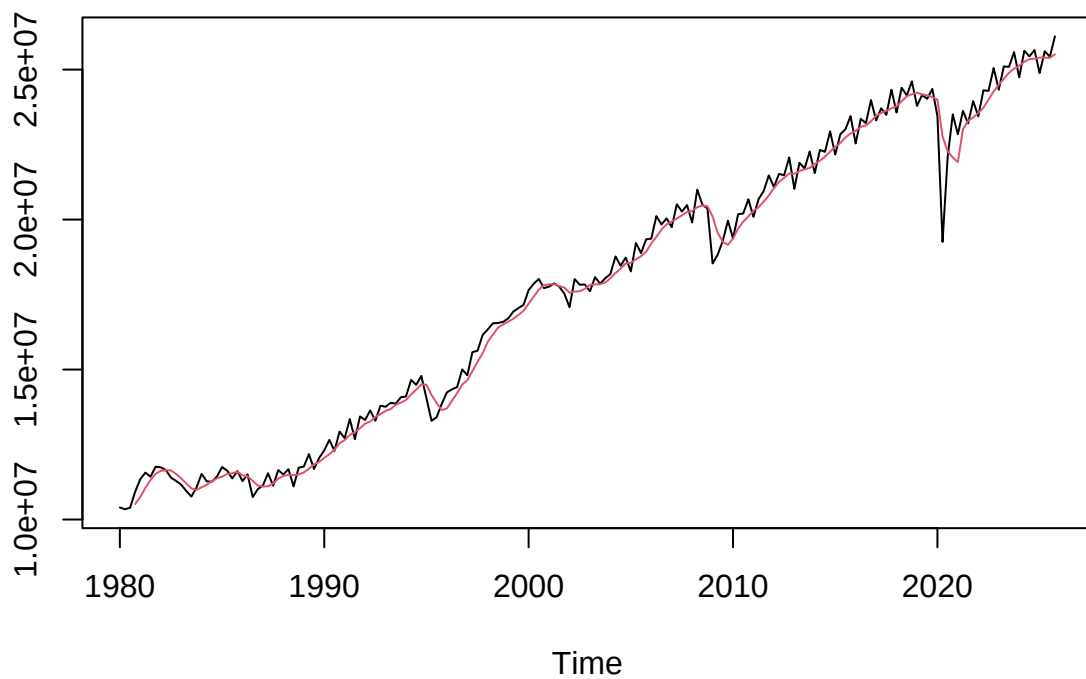
## Series: st2.ts
## ARIMA(1,1,1)
##
## Coefficients:
##          ar1          ma1
##       -0.1701   -0.2373
## s.e.    0.2330    0.2374
##
## sigma^2 = 3.394e+11:  log likelihood = -2688.11
## AIC=5382.21   AICc=5382.35   BIC=5391.84
ts.plot(st2.ts, st2.arima$fitted, col=1:2)
```



Modelo MA

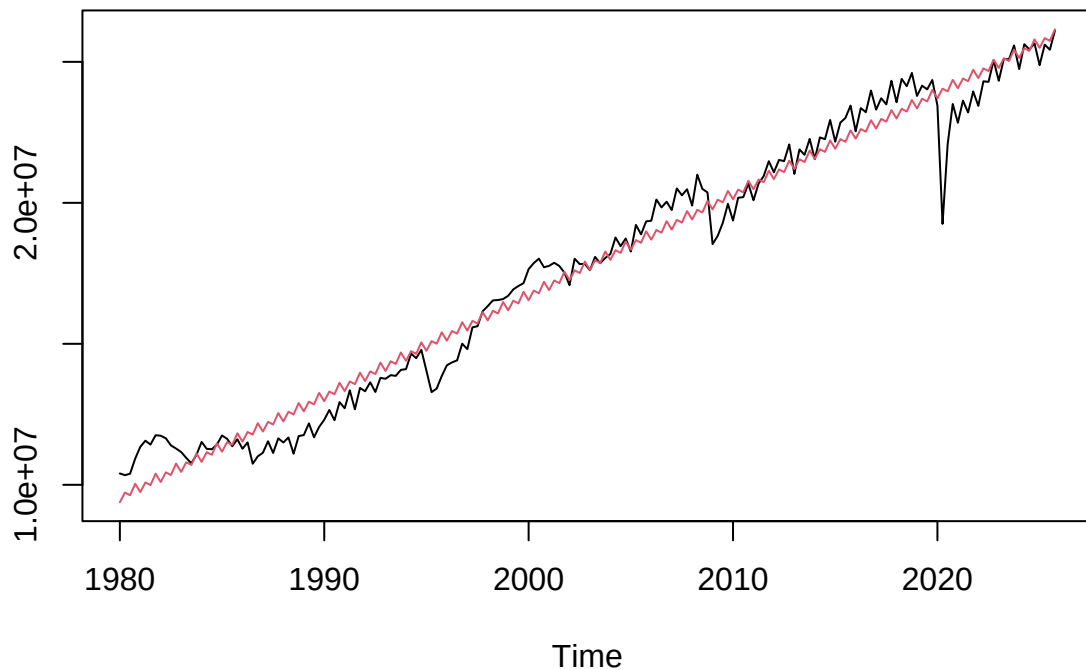
```
st2.sma <- SMA(st2.ts, n=4)

ts.plot(st2.ts, st2.sma, col=1:2)
```



Modelo MCS

```
st2.dummies <- seasonaldummy(st2.ts)
st2.mcs <- tslm(st2.ts ~ trend + season)
ts.plot(st2.ts, st2.mcs$fitted.values, col=1:2)
```



Pronósticos

Tenemos los pronósticos:

```
st2.arima.forecast <- forecast(st2.arima, h=6, level=95)
st2.sma.forecast <- forecast(st2.sma, h=6, level=95)
```

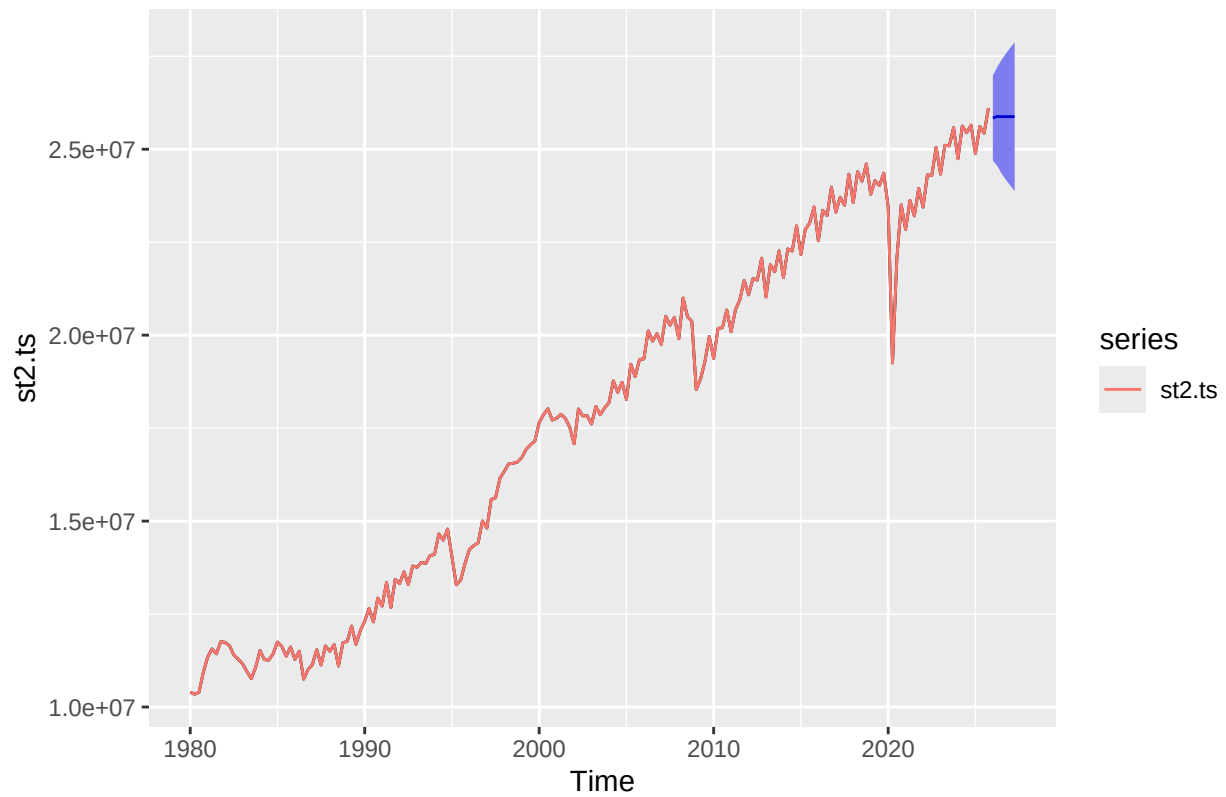
```
## Warning in lsfit(seq(maxn), head(y.sa, maxn)): 3 missing values deleted
## Warning in lsfit(seq(maxn), head(y.sa, maxn)): 3 missing values deleted
## Warning in lsfit(seq(maxn), head(y.sa, maxn)): 3 missing values deleted
## Warning in lsfit(seq(maxn), head(y.sa, maxn)): 3 missing values deleted
```

```
st2.mcs.forecast <- forecast(st2.mcs, h=6, level=95)
```

Comparación visual

```
autoplot(st2.arima.forecast) + autolayer(st2.ts)
```

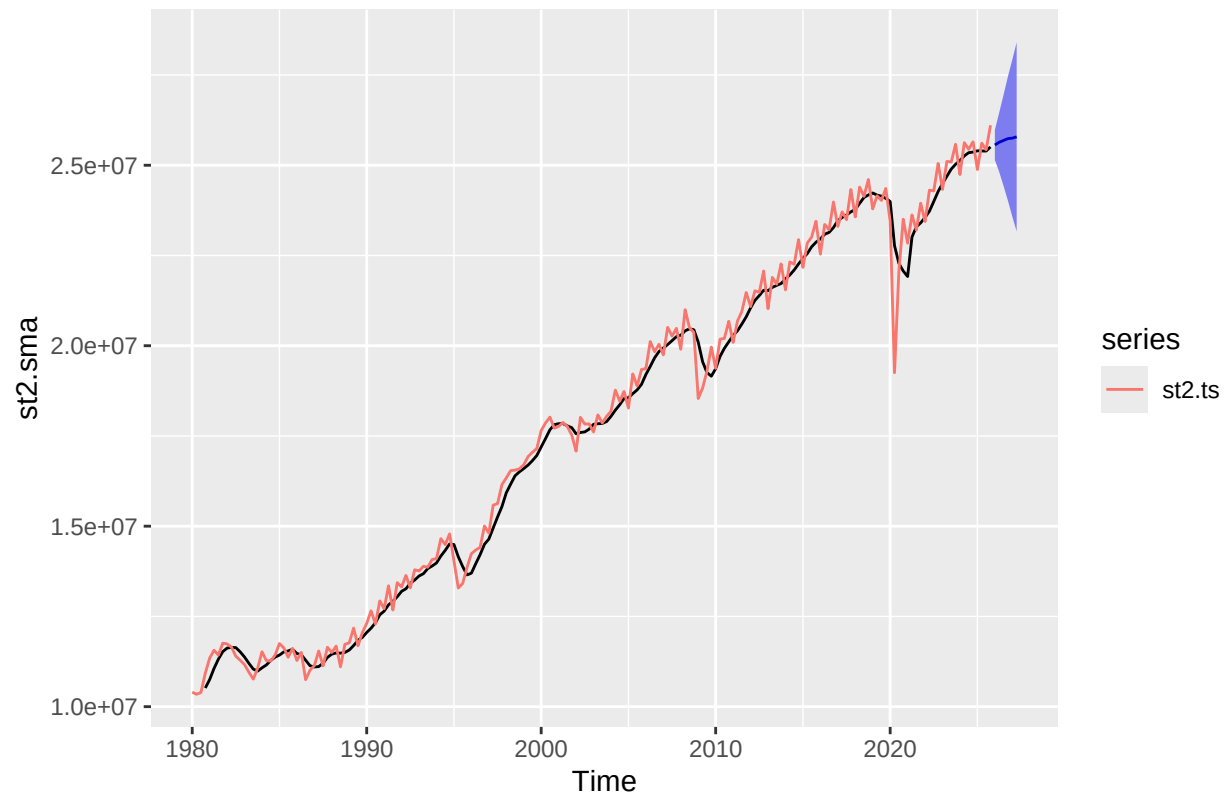
Forecasts from ARIMA(1,1,1)



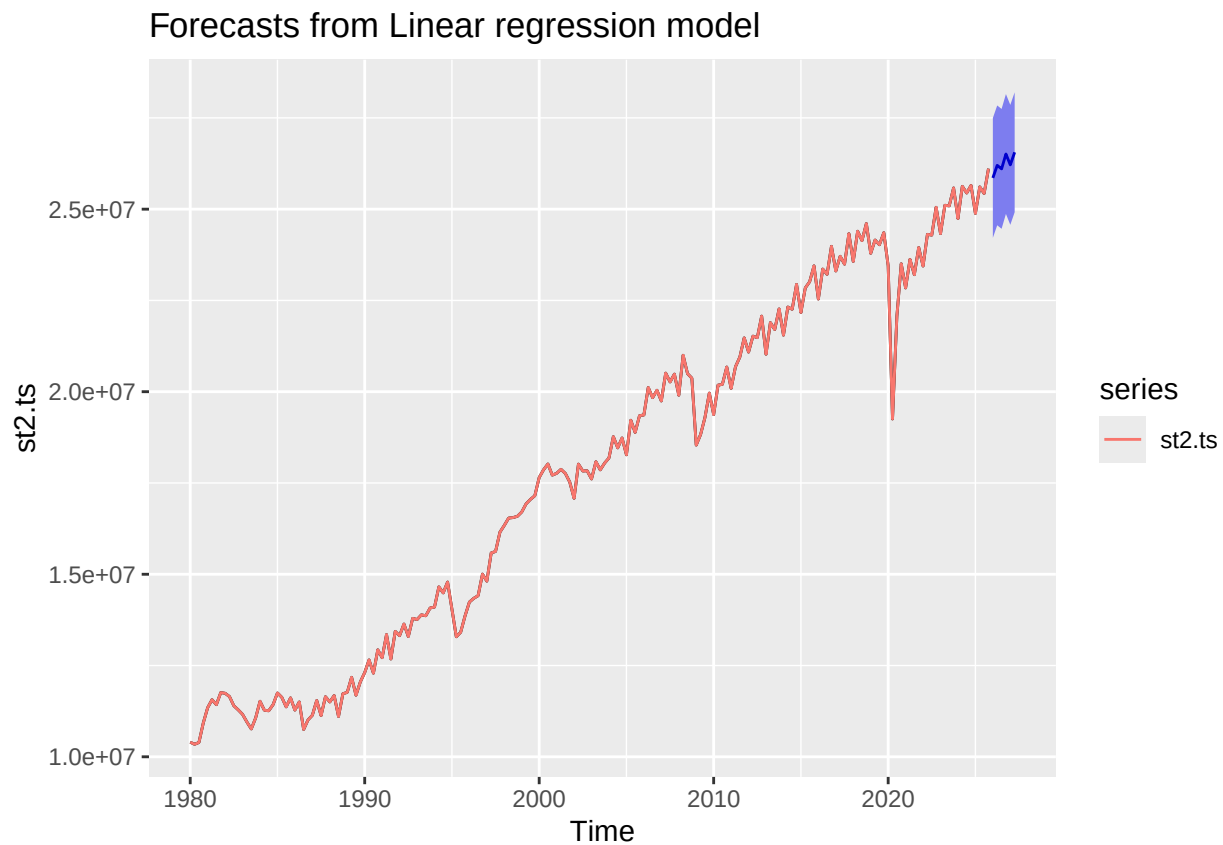
```
autoplot(st2.sma.forecast) + autolayer(st2.ts)
```

```
## Warning: Removed 3 rows containing missing values or values outside the scale range
## (`geom_line()`).
```


Forecasts from ETS(M,Ad,A)



```
autoplot(st2.mcs.forecast) + autolayer(st2.ts)
```



En este caso, el modelo MCS parece apegarse más a la serie y dar un mejor pronóstico. Los modelos ARIMA parece modelar bien la serie, pero con mal pronóstico; el modelo SMA no modela muy bien el comportamiento de la serie ni parece dar un buen pronóstico.